Toru Seo

Curriculum Vitae, as of November 5, 2026

Personal Information

Name: Toru SEO

Name in Japanese: 瀬尾 亨（セオ トオル）

Affiliation

Associate Professor, *Seo Lab., Department of Civil and Environmental Engineering, School of Environment and Society, Institute of Science Tokyo, Japan*

Contact

E-mail: seo.t.701d{at}m.isct.ac.jp; seo.t.aa{at}m.titech.ac.jp

Address: West Building 6 Room 411, 2-12-1-W6-10 Ookayama, Meguro-ku, Tokyo 152-8550, Japan

Personal Web page: https://toruseo.jp/index_en.html

Lab Web page: https://toruseo.jp/lab/en/

ORCID: 0000-0001-9767-7218

Web of Science ResearcherID: I-3748-2016

ResearchGate: https://www.researchgate.net/profile/Toru_Seo2

Google Scholar: https://scholar.google.com/citations?user=CAxkSpwAAAAJ

GitHub: https://github.com/toruseo

Degree

[2015-09-25] Doctor of Engineering, *Department of Civil Engineering, Graduate School of Science and Engineering, Tokyo Institute of Technology, Japan*

Working Experience

[2024-10/] Associate Professor, *Institute of Science Tokyo, Japan (change due to re-organization)*

[2021-04/2024-09] Associate Professor, *Tokyo Institute of Technology, Japan*

[2021-07/2022-03] Visiting Researcher, *Center for Spatial Information Science, The University of Tokyo, Japan*

[2018-06/2021-03] Assistant Professor, *The University of Tokyo, Japan*

[2016-04/2018-05] Research Staff, *Tokyo Institute of Technology, Japan*

[2017-08/2018-05] Research Associate, *University of Michigan, The United States*

[2017-06/2017-08] Visiting Scholar, *University of Michigan, The United States*

[2015-10/2016-03] Research Fellow (PD), *Japan Society for the Promotion of Science, Japan*

[2014-04/2015-09] Research Fellow (DC2), *Japan Society for the Promotion of Science, Japan*

Education

[2015-09] Dr.Eng., *Department of Civil Engineering, Graduate School of Science and Engineering, Tokyo Institute of Technology, Japan*

Doctoral dissertation: Traffic estimation with vehicles observing other vehicles

Supervisor: Prof. Yasuo Asakura

[2013-03] M.Eng., *Department of Civil Engineering, Graduate School of Science and Engineering, Tokyo Institute of Technology, Japan*
Master thesis: Traffic state estimation with Lagrangian observation (in Japanese)
Supervisors: Prof. Yasuo Asakura and Dr. Daisuke Fukuda

[2011-03] B.Eng., *Department of Civil and Environmental Engineering, School of Engineering, Tokyo Institute of Technology, Japan*
Graduation thesis: Pedestrian behavior modeling in a train station based on the concept of "Plan-Action" (in Japanese)
Supervisor: Dr. Daisuke Fukuda

Honors and Awards

[2026-09-01] Best Paper Award Shortlist (12 out of 754), *IEEE 29th International Conference on Intelligent Transportation Systems*

[2024-12-13] Best Poster Award, *The 22nd ITS Symposium 2024*

[2017-11-04] Outstanding Paper Award, *Committee of Infrastructure Planning and Management, Japan Society of Civil Engineers*

[2017-07] TRC Best Paper Award, *Transportation Research Part C: Emerging Technologies*

[2016-11-25] Kometani–Sasaki Prize (for Dissertation), *Institute of Systems Science Research*

[2016-06-20] The 30th Japan Society of Traffic Engineers Paper Award, *Japan Society of Traffic Engineers*

[2015-09-16] Best Paper Award, *IEEE 18th International Conference on Intelligent Transportation Systems*

[2014-08-08] Research Encouragement Award, *The 34th Conference of Japan Society of Traffic Engineers*

[2013-12-05] Outstanding Paper Award, *The 30th Japan Road Conference*

Competitive Research Funding

Principal Investigator

[2024-04/2028-03] KAKENHI Grant-in-Aid for Scientific Research (B), Japan Society for the Promotion of Science, *Development and Extensions of Normative Mathematical Models for Shared Autonomous Vehicle Transportation System*, PI: Toru Seo, Total budget: JPY 14200k.

[2024-06/2026-03] KAKENHI Grant-in-Aid for Challenging Research (Exploratory), Japan Society for the Promotion of Science, *Development of statistical analysis and modeling methodology for dynamic transportation phenomena based on directional statistics*, PI: Toru Seo, Total budget: JPY 4800k.

[2020-04/2024-03] KAKENHI Grant-in-Aid for Scientific Research (B), Japan Society for the Promotion of Science, *Estimation, prediction, and control of road transportation system by integrating traffic flow theory and machine learning*, PI: Toru Seo, Total budget: JPY 13500k.

[2016-04/2020-03] KAKENHI Grant-in-Aid for Young Scientists (B), Japan Society for the Promotion of Science, *Development and verification of model describing spatiotemporal dynamics of traffic flow consists of heterogeneous vehicles*, PI: Toru Seo, Total budget: JPY 3200k.

[2014-04/2016-03] KAKENHI Grant-in-Aid for JSPS Fellows, Japan Society for the Promotion of Science, *Road network traffic state estimation using information from cameras on probe vehicles*, PI: Toru Seo, Total budget: JPY 1700k.

Co-Investigator

[2026-08/2029-03] Research Fund by Committee on Advanced Road Technology, Ministry of Land, Infrastructure, Transport and Tourism, Japan, *Wide-area traffic state estimation using moving drones*, PI: Yosuke Kawasaki, Total budget: JPY 4500k.

[2026-04/2030-03] KAKENHI Grant-in-Aid for Scientific Research (B), Japan Society for the Promotion of Science, *Co-evolution of transportation and city in automated driving-era: evolutionary dynamical policy design based on integrated mathematical model*, PI: Koki Satsukawa, Total budget: JPY 2000k.

[2025-04/2030-03] Science and Technology Research Partnership for Sustainable Development (SATREPS),

Japan Science and Technology Agency and Japan International Cooperation Agency, *The Project for Data Driven Dynamic Transport Management in Emerging Metropolis for Climate Change Mitigation*, PI: Daisuke Fukuda.

[2025-04/2029-03] KAKENHI Grant-in-Aid for Scientific Research (A), Japan Society for the Promotion of Science, *Distributed evolutionary optimization and implementation of integrated transportation systems in the era of digital twin*, PI: Takamasa Iryo.

[2024-04/2028-03] KAKENHI Grant-in-Aid for Scientific Research (B), Japan Society for the Promotion of Science, *Development of real-time highway traffic accident risk mitigation system during transitional period for automated driving diffusion*, PI: Yasunori Muromachi, Total budget: JPY 1400k.

[2023-04/2026-03] KAKENHI Grant-in-Aid for Scientific Research (B), Japan Society for the Promotion of Science, *Validation of traffic flow theory on highway and strategic design of traffic control system*, PI: Kentaro Wada, Total budget: JPY 2200k.

[2021-04/2024-03] Research Fund by Committee on Advanced Road Technology, Ministry of Land, Infrastructure, Transport and Tourism, Japan, *Next-gen traffic measurement methods combining camera images and multiple observation data*, PI: Hideki Yaginuma, Total budget: JPY 0.

[2019-04/2022-03] KAKENHI Grant-in-Aid for Scientific Research (B), Japan Society for the Promotion of Science, *Traffic flow optimization by intervention control to automated vehicles*, PI: Yasuhiro Shiomi, Total budget: JPY 1800k.

[2019-04/2022-03] Research Fund by Committee on Advanced Road Technology, Ministry of Land, Infrastructure, Transport and Tourism, Japan, *Facility location and transportation management considering multi-scale modal cooperation*, PI: Takahiko Kusakabe, Total budget: JPY 1650k.

[2018-04/2021-03] Research Fund by Committee on Advanced Road Technology, Ministry of Land, Infrastructure, Transport and Tourism, Japan, *Tourism Congestion Management using Learning-based Monitoring and Prediction of Traffic*, PI: Takashi Fuse, Total budget: JPY 9000k.

[2017-04/2021-03] KAKENHI Grant-in-Aid for Scientific Research (A), Japan Society for the Promotion of Science, *Research on mathematical models for management of large-scale transportation network under mega disaster*, PI: Yasuo Asakura, Total budget: JPY 700k.

[2017-04/2020-03] KAKENHI Grant-in-Aid for Scientific Research (B), Japan Society for the Promotion of Science, *Integrated Analytical Modeling of Urban Rail Transit System with High-Frequent Operations*, PI: Daisuke Fukuda, Total budget: JPY 1200k.

[2017-04/2019-03] KAKENHI Grant-in-Aid for Challenging Research (Exploratory), Japan Society for the Promotion of Science, *Market diffusion and social influences of automated cars: Integrated approach from traffic engineering and transport economics*, PI: Daisuke Fukuda, Total budget: JPY 600k.

Affiliated Societies

IEEE, IEEE Intelligent Transportation Systems Society

Japan Society of Civil Engineers

Japan Society of Traffic Engineers

Information Processing Society of Japan

Academic Services

Editor

[2026/] Associate editor, Japanese Journal of JSCE, Special Issue (Infrastructure Planning and Management (Method and Technology))

[2025/] Associate editor, International Journal of Intelligent Transportation Systems Research

[2024/] Editorial committee member, JSTE Journal of Traffic Engineering

[2022/2025] Associate editor, Japanese Journal of JSCE

[2017] Guest editor, JSTE Traffic Engineering, Vol.52, No.4

Journal reviewer

Transportation Research Part A: Policy and Practice; Transportation Research Part B: Methodological; Transportation Research Part C: Emerging Technologies; Transportation Research Part E: Logistics and Transportation Review; Transportation Science; Transportation; Transportation Research Record; Transportmetrica A; Transportmetrica B; IEEE Transactions on Intelligent Transportation Systems; IEEE Intelligent Transportation Systems Magazine; IEEE Transactions on Vehicular Technology; IEEE Transactions on Systems, Man, and Cybernetics; IET Intelligent Transportation System; Travel Behaviour and Society; Transportation Letters; Transportation Planning and Technology; International Journal of Sustainable Transportation; Physica A: Statistical Mechanics and its Applications; International Journal of Intelligent Transportation Systems Research; Asian Transportation Studies; IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences; Engineering Applications of Artificial Intelligence; Artificial Life and Robotics; Hydrological Research Letters; Sensors; PLOS ONE; Entropy; Energies; MethodsX; Journal of Japan Society of Civil Engineers, Ser. D3; JSTE Journal of Traffic Engineering; Transport Policy Studies' Review

International conference session chair

IEEE International Conference on Intelligent Transportation Systems; EURO Working Group on Transportation Meeting; Triennial Symposium on Transportation Analysis conference (TRISTAN)

Committee member etc.

[2026-06-01/] Member, Scientific subcommittee, Committee of Infrastructure Planning and Management, Japan Society of Civil Engineers

[2025-08/] Member, Committee on research on traffic management on highway, Japan Society of Traffic Engineers

[2025-04-02/] Associate Editor, Editorial Board, International Journal of Intelligent Transportation Systems Research, Springer/ITS Japan

[2024-04-01/] Member, Second academic subcommittee, Japan Society of Traffic Engineers

[2023-06/] Secretary, Secretariat of committee on traffic technology, Hanshin Expressway Company Limited

[2022-08-04/] Member, Study group for rail transport project evaluation, Japan Transport and Tourism Research Institute

[2024-03-05/2025-06-27] Member, Scientific committee, The 12th Triennial Symposium on Transportation Analysis conference (TRISTAN XII)

[2022-06-10/2025-06] Associate Editor, Editorial committee on Journal of Japan Society of Civil Engineers, Japan Society of Civil Engineers

[2022-06-08/] Member, Committee on novel traffic control methods, Japan Society of Traffic Engineers

[2022-04-01/2025-05-31] Member, Committee on practical application of theory on traffic congestion at highway basic segments, Japan Society of Traffic Engineers

[2022-04/2023-01] Program Committee Member, The 20th ITS Symposium 2022, ITS Japan

[2018-10/2021-03] Member, Study group for rail transport project evaluation, Japan Transport and Tourism Research Institute

[2018-09/2019-06] Member, Working group for advanced road management using vehicle trajectory data, Japan Society of Traffic Engineers

[2017-02/2017-10] Invited Member, "Traffic Engineering" editorial committee, Japan Society of Traffic Engineers

Teaching Experience

Classes

[2025/] Frontiers in Civil Engineering (partial), *Institute of Science Tokyo (Graduate school)*

[2025/] Basic Mathematics for System Science (partial), *Institute of Science Tokyo*

[2025/] School of Environment and Society Academic Group Basic Science I (partial), *Institute of Science Tokyo*

[2025/] Traffic and Transportation Systems, *Institute of Science Tokyo*

[2025/] Integrated Civil and Environmental Engineering Project (partial), *Institute of Science Tokyo*

[2025/] Urban and Transportation Planning Project (partial), *Institute of Science Tokyo*

[2024/] Transportation Science and Simulation, *Institute of Science Tokyo (Graduate school)*

[2024/] Research Opportunity in Laboratories (CVE) (partial), *Institute of Science Tokyo*

[2021/2023] Transportation Science and Simulation, *Tokyo Institute of Technology (Graduate school)*

[2021/2023] Frontiers in Civil Engineering (partial), *Tokyo Institute of Technology (Graduate school)*

[2024] Basic Mathematics for System Science (partial), *Tokyo Institute of Technology*

[2023/2024] School of Environment and Society Academic Group Basic Science I (partial), *Tokyo Institute of Technology*

[2023/2024] Introduction to Infrastructure and Environment (partial), *Tokyo Institute of Technology*

[2022/2024] Traffic and Transportation Systems, *Tokyo Institute of Technology*

[2021/2023] Research Opportunity in Laboratories (CVE) (partial), *Tokyo Institute of Technology*

[2021/2022, 2024] Integrated Civil and Environmental Engineering Project (partial), *Tokyo Institute of Technology*

[2021/2024] Urban and Transportation Planning Project (partial), *Tokyo Institute of Technology*

[2021/2023] Surveying (partial, primary), *Tokyo Institute of Technology*

[2021/2022] Fundamentals of Infrastructure Planning (partial, primary), *Tokyo Institute of Technology*

[2019/2020] Spatial Information Engineering I (partial), *The University of Tokyo*

[2018/2019] Applied Project III (partial), *The University of Tokyo*

[2018/2020] Fieldwork Exercise (partial), *The University of Tokyo*

[2018/2020] Fieldwork in Spatial Information Engineering (partial), *The University of Tokyo*

[2011] Infrastructure Planning and Design (teaching assistant), *Tokyo Institute of Technology*

Supervision

Doctor: 2 (Tokyo Institute of Technology)

- Nagasaki, Kota. Transportation systems analysis with angle: Modeling and empirical studies, Doctor of Engineering, Tokyo Institute of Technology, 2024-03-26
- Dahiya, Garima. Fundamental diagrams and traffic state estimation methods: analysis and modeling using Zen Traffic Data, Doctor of Engineering, Tokyo Institute of Technology, 2022-09-22

Master: 18 (Institute of Science Tokyo: 10, Tokyo Institute of Technology: 8)

Bachelor: 10 (Institute of Science Tokyo: 3, Tokyo Institute of Technology: 7)

Master (auxiliary): 2 (The University of Tokyo)

Bachelor (auxiliary): 3 (The University of Tokyo)

Awards won by supervised students: Paper Award, the Japan Society of Photogrammetry and Remote Sensing Reiwa 1st Fall Conference (2019); Kimura Award, Department of Civil and Environmental Engineering, Tokyo Institute of Technology (2022); Kikkawa–Yamaguchi Award, Department of Civil and Environmental Engineering, Tokyo Institute of Technology and its Alumni Association “Kyuyu” (2022); Outstanding Master Thesis Award, Department of Civil and Environmental Engineering, Tokyo Institute of Technology (2024); Student Award, Independent Research Project Presentation, Department of Civil and Environmental Engineering, Institute of Science Tokyo (2025); Kimura Award, Department of Civil and Environmental Engineering, Institute of Science Tokyo (2026)

Publications

Book

1. Seo, T. *Macroscopic Traffic Flow Simulation: Fundamental Mathematical Theory and Python Implementation*. Corona Publishing Co., Ltd., 2023. (in Japanese)

Book Chapters

2. Wada, K., Seo, T., and Shiomi, Y. Flow breakdown. In Vickerman, R. (Ed.), *International Encyclopedia of Transportation*, Vol. 4, pp. 143–153. Elsevier, 2021
1. Wada, K., Seo, T., and Shiomi, Y. Bottleneck. In Vickerman, R. (Ed.), *International Encyclopedia of Transportation*, Vol. 4, pp. 134–142. Elsevier, 2021

Refereed International Journal Articles

18. Seo, T., Kusakabe, T., and Oguchi, T. Estimation of jam density and saturation flow at signalized intersections using only low penetration rate connected vehicle GNSS trajectories: Case study using actual data in Tokyo. *International Journal of Intelligent Transportation Systems Research*, in press
17. Ishii, Y., Kawase, R., and Seo, T. Multi-objective optimization of integrated freight and passenger transportation in shared autonomous vehicle systems. *International Journal of Intelligent Transportation Systems Research*, Vol. 23, pp. 908–923, 2025
16. Seo, T. UXsim: lightweight mesoscopic traffic flow simulator in pure Python. *Journal of Open Source Software*, Vol. 10, No. 106, p. 7617, 2025
15. Iizuka, K. and Seo, T. Passenger-oriented distributed traffic signal control using dynamic programming with vehicle queue spillback and waiting time constraints. *International Journal of Intelligent Transportation Systems Research*, Vol. 22, pp. 579–602, 2024
14. Chen, X., Qin, G., Seo, T., Yin, J., Tian, Y., and Sun, J. A macro-micro approach to reconstructing vehicle trajectories on multi-lane freeways with lane changing. *Transportation Research Part C: Emerging Technologies*, Vol. 160, p. 104534, 2024
13. Maruyama, R. and Seo, T. Integrated public transportation system with shared autonomous vehicles and fixed-route transits: Dynamic traffic assignment-based model with multi-objective optimization. *International Journal of Intelligent Transportation Systems Research*, Vol. 21, pp. 99–114, 2023
12. Seo, T., Wada, K., and Fukuda, D. Fundamental diagram of urban rail transit considering train–passenger interaction. *Transportation*, Vol. 50, pp. 1399–1424, 2023
11. Seo, T. and Asakura, Y. Multi-objective linear optimization problem for strategic planning of shared autonomous vehicle operation and infrastructure design. *IEEE Transactions on Intelligent Transportation Systems*, Vol. 23, pp. 3816–3828, 2022
10. Seo, T., Kawasaki, Y., Kusakabe, T., and Asakura, Y. Fundamental diagram estimation by using trajectories of probe vehicles. *Transportation Research Part B: Methodological*, Vol. 122, pp. 40–56, 2019
9. Thaithatkul, P., Seo, T., Kusakabe, T., and Asakura, Y. Evolution of a dynamic ridesharing system based on rational behaviour of users. *International Journal of Sustainable Transportation*, Vol. 13, No. 8, pp. 614–626, 2019
8. Seo, T., Kusakabe, T., Gotoh, H., and Asakura, Y. Interactive online machine learning approach for activity-travel survey. *Transportation Research Part B: Methodological*, Vol. 123, pp. 362–373, 2019 (Selected paper from IATBR2015)
7. Lykov, S., Seo, T., and Asakura, Y. Analysis of spatiotemporal dependencies in two-dimensional traffic flow in large-scale urban area with probe vehicle data. *Journal of the Eastern Asia Society for Transportation Studies*, Vol. 12, pp. 1676–1696, 2017
6. Seo, T., Bayen, A. M., Kusakabe, T., and Asakura, Y. Traffic state estimation on highway: A comprehensive survey. *Annual Reviews in Control*, Vol. 43, pp. 128–151, 2017
5. Thaithatkul, P., Seo, T., Kusakabe, T., and Asakura, Y. A passengers matching problem in ridesharing systems by considering user preference. *Journal of the Eastern Asia Society for Transportation Studies*, Vol. 11, pp. 1416–1432, 2015

4. Seo, T. and Kusakabe, T. Probe vehicle-based traffic state estimation method with spacing information and conservation law. *Transportation Research Part C: Emerging Technologies*, Vol. 59, pp. 391–403, 2015 (Selected paper from ISTTT21)
3. Seo, T., Kusakabe, T., and Asakura, Y. Estimation of flow and density using probe vehicles with spacing measurement equipment. *Transportation Research Part C: Emerging Technologies*, Vol. 53, pp. 134–150, 2015 **[Best Paper Award]**
2. Fukuda, D., Seo, T., Yamada, K., Yaginuma, H., and Matsuyama, N. An econometric-based model of pedestrian walking behavior implicitly considering strategic or tactical decisions. In Weidmann, U., Kirsch, U., and Schreckenberg, M. (Eds.), *Pedestrian and Evacuation Dynamics 2012*, pp. 615–624. Springer International Publishing, 2014
1. Narioka, N., Seo, T., Kusakabe, T., and Asakura, Y. Incident detection method using longitudinal occupancy time-series data. *Journal of the Eastern Asia Society for Transportation Studies*, Vol. 10, pp. 1720–1733, 2013

Refereed Japanese Journal Articles

15. Seo, T., Kusakabe, T., Shiomi, Y., and Oguchi, T. Estimation of saturation flow at signalized intersections using high-resolution connected vehicle trajectory data with low penetration rate. *JSTE Journal of Traffic Engineering*, Vol. 12, No. 3, pp. 12–26, 2026
14. Nagasaki, K. and Seo, T. Traffic congestion analysis with directional statistics: Descriptive model by angular distributions for road network and OD demand. *Japanese Journal of JSCE*, Vol. 81, No. 9, p. 2400194, 2025. (in Japanese)
13. Hayashi, K. and Seo, T. Vehicle and section-dependent speed-spacing relation estimation using hierarchical Bayesian model. *Japanese Journal of JSCE*, Vol. 80, No. 20, p. 2420029, 2024. (in Japanese)
12. Oda, K., Seo, T., and Nakanishi, W. Estimation of location-dependent fundamental diagrambased on connected vehicle data and sparse modeling. *JSTE Journal of Traffic Engineering*, Vol. 10, No. 1, pp. A_316–A_323, 2024. (in Japanese)
11. Nagasaki, K. and Seo, T. Route choice model based on angular indicator and case study. *Japanese Journal of JSCE*, Vol. 79, No. 20, p. 2320059, 2023. (in Japanese)
10. Sato, K., Seo, T., and Fuse, T. Data-driven dynamic congestion toll optimization methods based on reinforcement learning. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)*, Vol. 76, No. 5, pp. I_1273–I_1285, 2021. (in Japanese)
9. Seo, T. and Kusakabe, T. Traffic state estimation using satellite remote sensing and probe vehicles. *JSTE Journal of Traffic Engineering*, Vol. 5, No. 2, pp. A_1–A_10, 2019. (in Japanese)
8. Aiko, S., Thaithatkul, P., Seo, T., and Asakura, Y. Optimum routing of ride share vehicles for given activity patterns. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)*, Vol. 73, No. 5, pp. I_1233–I_1242, 2017. (in Japanese)
7. Wada, K., Seo, T., Nakanishi, W., Satsukawa, K., and Yanagihara, M. Recent advances in kinematic wave theory of traffic flows: variational formulation and network extension. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)*, Vol. 73, No. 5, pp. I_1139–I_1158, 2017. (in Japanese)
6. Fukuda, D., Mizuguchi, M., Seo, T., Kusakabe, T., and Asakura, Y. Evaluation of area level travel time reliability using large-scale probe vehicle trajectories recorded for a long period. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)*, Vol. 73, No. 5, pp. I_1105–I_1118, 2017. (in Japanese)
5. Seo, T., Kusakabe, T., and Asakura, Y. Trip purpose estimation method for probe person survey using sequential learning. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)*, Vol. 73, No. 5, pp. I_517–I_526, 2017. (in Japanese) **[Outstanding Paper Award]**
4. Seo, T., Kusakabe, T., and Asakura, Y. Methodology for calibration of fundamental diagram based on trajectories of sampled vehicles: Concept and numerical experiment. *JSTE Journal of Traffic Engineering*, Vol. 2, No. 2, pp. A_1–A_10, 2016. (in Japanese)

3. Narioka, N., Seo, T., Kusakabe, T., and Asakura, Y. A method for detecting incidents from traffic detector data based on the non-parametric statistics. *JSTE Journal of Traffic Engineering*, Vol. 1, No. 1, pp. 11–20, 2015. (in Japanese) **[JSTE Paper Award]**
2. Seo, T., Kusakabe, T., and Asakura, Y. Estimation of traffic state using probe vehicles that equipped with spacing measurement devices. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)*, Vol. 69, No. 5, pp. I_809–I_818, 2013. (in Japanese)
1. Seo, T., Yaginuma, H., and Fukuda, D. Modeling pedestrian behavior based on the concept of "Plan-Action" structure: An application in a train station. *Journal of Japan Society of Civil Engineers, Ser. D3 (Infrastructure Planning and Management)*, Vol. 68, No. 5, pp. I_679–I_690, 2012. (in Japanese)

Refereed International Conference Presentations

55. Li, S., Seo, T., and Wada, K. Data assimilation-based estimation of input variables for generalized bathtub model: A case study of Tokyo Metropolitan Expressway. In *IEEE 29th International Conference on Intelligent Transportation Systems*, Naples, Italy, 2026
54. Koseki, S., Kawase, R., and Seo, T. Wasserstein distance-based rolling horizon control for demand responsive transport under uncertainty. In *IEEE 29th International Conference on Intelligent Transportation Systems*, Naples, Italy, 2026 **[Best Paper Award Shortlist]**
53. Ohkawa, Y., Seo, T., and Kawase, R. Estimation–control integrated method for mobile data-driven area-wide traffic management. In *IEEE 29th International Conference on Intelligent Transportation Systems*, Naples, Italy, 2026
52. Kawase, R., Satsukawa, K., and Seo, T. Flexible and reliable network design for emerging transportation services: Multi-stage stochastic programming approach. In *The 26th International Symposium on Transportation and Traffic Theory (ISTTT)*, 2026
51. Ma, Y. and Seo, T. Incorporating graph learning and attention mechanisms into route choice models to capture local and global correlations. In *Proceedings of the 29th International Conference of Hong Kong Society for Transportation Studies*, Hong Kong, China, 2025
50. Oda, K. and Seo, T. Passenger flow estimation method for ridesharing systems using inverse problem. In *IEEE 28th International Conference on Intelligent Transportation Systems*, pp. 2504–2510, Gold Coast, Australia, 2025
49. Zhong, H. and Seo, T. A macroscopic comparison method for road network datasets using geographical traffic distribution. In *International Symposium on Transportation Data and Modelling 2025*, Montreal, Canada, 2025
48. Seo, T. and Kawase, R. Facility and dynamic fare design for multimodal automated vehicle logistics system under traffic flow constraints. In *The 12th Triennial Symposium on Transportation Analysis conference (TRISTAN XII)*, Okinawa, Japan, 2025
47. Hayashi, K. and Seo, T. Estimating source of traffic congestion from vehicle trajectory data using hierarchical Bayesian model. In *Proceedings of the 28th International Conference of Hong Kong Society for Transportation Studies*, Hong Kong, 2024
46. Ma, Y. and Seo, T. Incorporating graph neural network into route choice model in road network. In *IEEE 27th International Conference on Intelligent Transportation Systems*, pp. 603–608, Edmonton, Canada, 2024
45. Seo, T., Maruyama, R., Wada, K., and Zhou, Y. Dynamic system optimal pricing for shared autonomous vehicles in congestible networks: Theoretical properties. In *Conference in Emerging Technologies in Transportation Systems (TRC-30)*, Crete, Greece, 2024
44. Fujiya, K., Nagasaki, K., and Seo, T. Modeling pedestrian fundamental diagram based on directional statistics. In *Proceedings of the 27th International Conference of Hong Kong Society for Transportation Studies*, Hong Kong, 2023
43. Nagasaki, K. and Seo, T. Route choice model using angular indicators. In *Transportation Research Procedia*, Vol. 78, pp. 238–245, 2024. (The 25th EURO Working Group on Transportation Meeting, 6–8 September 2023, Santander, Spain)

42. Zhong, H., Seo, T., Nakanishi, W., Yasuda, S., Asakura, Y., and Iryo, T. Generation of aggregated road network by vehicle trajectory data. In *International Symposium on Transportation Data and Modelling 2023*, Ispra, Italy, 2023
41. Seo, T. Understanding large-scale traffic flow using model-based and data-driven dimension reduction: with COVID-19 and Olympic-Paralympic case study. In *International Symposium on Transportation Data and Modelling 2023*, Ispra, Italy, 2023
40. Maruyama, R. and Seo, T. Dynamic user optimal model for shared autonomous vehicles system: Development and systematic comparison with social optimal model. In *IEEE 25th International Conference on Intelligent Transportation Systems*, Web conference, 2022
39. Seo, T. and Asakura, Y. Multi-objective linear optimization for strategic planning of shared autonomous vehicle operation and infrastructure design. In *The 8th International Symposium on Dynamic Traffic Assignment*, Web conference (Postponed from 2020), 2021
38. Seo, T., Tago, Y., Shinkai, N., Nakanishi, M., Tanabe, J., Ushirogochi, D., Kanamori, S., Abe, A., Kodama, T., Yoshimura, S., Ishihara, M., and Nakanishi, W. Evaluation of large-scale complete vehicle trajectories dataset on two kilometers highway segment for one hour duration: Zen Traffic Data. In *2021 International Symposium on Transportation Data and Modelling*, Web conference (Postponed from 2020), 2021
37. Sato, K., Seo, T., and Fuse, T. A reinforcement learning-based dynamic congestion pricing method for the morning commute problems. In *Transportation Research Procedia*, Vol. 52, pp. 347–355, 2021. (The 23rd EURO Working Group on Transportation Meeting, 16–18 September 2020, Web conference)
36. Seo, T. Calibration-free traffic state estimation method using single detector and connected vehicles with Kalman filtering and RTS smoothing. In *IEEE 23rd International Conference on Intelligent Transportation Systems*, Web conference, 2020
35. Sakai, K., Seo, T., and Fuse, T. Traffic density estimation method from small satellite imagery: Towards frequent remote sensing of car traffic. In *IEEE 22nd International Conference on Intelligent Transportation Systems*, pp. 1776–1781, Auckland, New Zealand, 2019
34. Seo, T. Trial-and-error congestion pricing for morning commute problem with day-to-day dynamics. *Transportation Research Procedia*, Vol. 47, pp. 561–568, 2020. (The 22nd EURO Working Group on Transportation Meeting, 18–20 September 2019, Barcelona, Spain)
33. Fukuda, D., Imaoka, M., and Seo, T. Empirical investigation of fundamental diagram for urban rail transit using Tokyo's commuter rail data. In *TRANSITDATA2019: 5th International Workshop and Symposium*, Paris, France, 2019
32. Seo, T. and Yin, Y. Optimal pricing for departure time choice problems with unknown preference and demand: Trial-and-error approach. In *Transportation Research Board 98th Annual Meeting*, 2019
31. Seo, T. and Kusakabe, T. Use of small satellites and connected vehicles for large-scale traffic monitoring in road network. In *IEEE 21st International Conference on Intelligent Transportation Systems*, pp. 2805–2810, Maui, The United States, 2018
30. Thaithatkul, P., Seo, T., Kusakabe, T., and Asakura, Y. Adoption of dynamic ridesharing system under influence of information on social network. *Transportation Research Procedia*, Vol. 37, pp. 401–408, 2019. (The 21st EURO Working Group on Transportation Meeting, 17–19 September 2018, Braunschweig, Germany)
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